Microsoft Fabric Warehouse

Overview & Architecture

DBA Technical Reference | Volume 1 of 9

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1. What is Microsoft Fabric?

Microsoft Fabric is an end-to-end analytics platform delivered as a Software-as-a-Service (SaaS) solution on Azure. It consolidates multiple data and analytics workloads into a single unified platform, eliminating the need for separate cloud services and complex integrations.

Fabric brings together the following experiences:

- Data Engineering (Spark-based notebooks and pipelines)
- Data Factory (data movement and transformation pipelines)
- Data Warehouse (enterprise SQL analytics)
- Data Science (ML model development and deployment)
- Real-Time Intelligence (streaming and event processing)
- Power BI (reporting and visualization)

INFO: All Fabric experiences share a single storage layer called OneLake and a unified identity, governance, and billing model — making it a truly integrated platform.

2. What is Fabric Warehouse?

Fabric Warehouse is the enterprise SQL data warehouse experience within Microsoft Fabric. It is purpose-built for large-scale analytical workloads and supports full transactional T-SQL capabilities including DML, DDL, stored procedures, views, and functions.

Key characteristics of Fabric Warehouse:

- Full T-SQL support: SELECT, INSERT, UPDATE, DELETE, MERGE, stored procedures, views, TVFs
- Star schema and dimensional modeling support
- Automatic statistics management and query optimization
- Seamless connectivity via SSMS, Azure Data Studio, and Power BI
- Built on open Delta Lake and Parquet file formats
- No infrastructure management — fully managed by Microsoft

3. SaaS Architecture Model

Unlike traditional on-premises SQL Server or even Azure SQL Database (PaaS), Fabric Warehouse is a fully managed SaaS offering. This has significant implications for the DBA role:

Responsibility	Traditional DBA	Fabric DBA
OS & Server Management	Full ownership	Not required — managed by Microsoft
Database Engine Upgrades	Manual patching	Automatic — no action needed
Storage Management	Disk provisioning	Automatic — OneLake scales automatically
Compute Scaling	CPU/RAM configuration	Adjust Capacity Units (CUs)
Backup & Recovery	Configure & schedule	Restore-in-place via portal
Security	Configure at OS+DB level	Workspace + Object level permissions
Query Tuning	Full control	Statistics, query design, DMV monitoring

4. Storage Layer: Delta Lake & OneLake

All data in Fabric Warehouse is stored as Delta Lake tables using the Parquet columnar file format within OneLake — Microsoft's unified, logical data lake.

Delta Lake key features relevant to DBAs:

- ACID Transactions: Atomicity, Consistency, Isolation (Snapshot), Durability — guaranteed at table level
- Time Travel: Delta log maintains historical versions of data for auditing and recovery
- Schema Enforcement: Prevents corrupt or incompatible data from being written
- Immutable Files: Parquet files are never modified in place — INSERTs create new files, UPDATEs/DELETEs create new versions
- Compaction: Background service periodically compacts small Parquet files for read performance

TIP: Because Parquet files are immutable, INSERT operations almost never cause lock conflicts. However, concurrent UPDATE and DELETE statements on the same table can conflict — design ETL accordingly.

5. Compute Layer: Capacity Units (CUs)

Fabric compute is measured and billed in Capacity Units (CUs) rather than traditional vCores or DTUs. A Fabric Capacity is provisioned at the tenant or workspace level and shared across all workloads running within it.

Capacity SKU	CUs	Recommended Use
F2	2	Development and testing
F4	4	Small workloads and POCs
F8	8	Small production workloads
F16	16	Medium production workloads
F32	32	Large data warehouses
F64	64	Enterprise-scale analytics
F128	128	Very large enterprise workloads
F256+	256+	Mission-critical, high concurrency

INFO: Fabric supports Burstable Capacity — workloads can temporarily consume more CUs than provisioned, accumulating a 'CU debt' that is repaid during idle periods.

6. SQL Analytics Endpoint

Every Fabric Warehouse (and Lakehouse) automatically provisions a SQL Analytics Endpoint. This is a read-only T-SQL interface that exposes the underlying Delta Lake tables without requiring data movement.

SQL Analytics Endpoint capabilities:

- Query Delta Lake tables using standard T-SQL SELECT statements
- Create views, inline table-valued functions (TVFs), and stored procedures
- Define column-level and row-level security policies
- Connect using SSMS, Azure Data Studio, or any ODBC/JDBC client
- All workspaces share a single SQL Endpoint URL — workspace acts as server, warehouses appear as databases

WARNING: The SQL Analytics Endpoint is read-only for Lakehouse tables. To write data, use the Warehouse SQL editor or T-SQL DML against the Warehouse directly.

7. Fabric Workloads & Integration

Fabric Warehouse integrates natively with all other Fabric experiences, enabling end-to-end data pipelines without data copying:

Fabric Experience	Integration with Warehouse
Data Factory Pipelines	Load and transform data directly into Warehouse tables via Copy Activity or T-SQL
Dataflows Gen2	Write transformed data from Power Query into Warehouse tables
Lakehouse	Query Lakehouse tables directly from Warehouse using cross-database T-SQL
Power BI	Auto-generated semantic model from Warehouse schema for instant reporting
Data Science (Notebooks)	Spark notebooks can read/write Warehouse data via OneLake APIs
Real-Time Intelligence	Stream processed events into Warehouse for historical storage

8. Warehouse vs Lakehouse vs SQL Database

Feature	Fabric Warehouse	Fabric Lakehouse	SQL Database in Fabric
Primary Use	SQL analytics & DW	Big data & Data Lake	OLTP & app database
Write Interface	T-SQL DML	Spark / Files / T-SQL	T-SQL DML
Read Interface	T-SQL	T-SQL + Spark	T-SQL
Storage Format	Delta Lake / Parquet	Delta Lake / Parquet	Row-based pages
Transactions	Snapshot Isolation	Limited	Full ACID
Best For	Star schemas, BI	Data exploration, ML	Applications, OLTP

9. Key DBA Concepts & Terminology

Term	Definition
OneLake	Microsoft's unified logical data lake — single storage for all Fabric workloads
Delta Lake	Open-source storage layer adding ACID transactions to data lake storage
Parquet	Columnar binary file format used for efficient analytical query performance
Capacity Unit (CU)	Unit of compute and billing in Fabric — replaces vCores/DTUs
Workspace	Organizational container in Fabric — equivalent to a database server in SSMS
SQL Analytics Endpoint	Auto-provisioned T-SQL read interface for Lakehouses and Warehouses
DMV	Dynamic Management View — system view exposing engine operational state
Query Insights	Auto-created schema in Warehouse providing historical query analytics
Snapshot Isolation	Default and only isolation level in Fabric Warehouse
BCDR	Business Continuity and Disaster Recovery — Fabric's regional failover capability